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Batch 15

Algorithm

1. Start.
2. Read `size`.
3. Declare the variable-length array `nums[size]`.
4. Read all the elements of `nums[]`.
5. Pass the address of `nums[]` to `duplicate_element()`.
6. Initialize `result = 0`.
7. XOR all elements of the array with `result`.
8. XOR all integers from `1` to `size - 1` with `result`.
9. Because `x ^ x = 0`, all non-repeated values cancel out.
10. The remaining value is the repeated element.
11. Display the repeated element.
12. Stop.

Test Case Table

Test	Input Array	Size	Expected Output
1	`1 3 4 2 2`	5	`Repeated element: 2`
2	`3 1 3 4 2`	5	`Repeated element: 3`
3	`1 4 5 2 3 4`	6	`Repeated element: 4`
4	`1 2 3 4 5 5`	6	`Repeated element: 5`
5	`1 1`	2	`Repeated element: 1`

C program

```
#include <stdio.h>

void duplicate_element(int *nums, int size)
{
    int result = 0;
```

```
    // XOR all array elements
    for (int i = 0; i < size; i++)
    {
        result ^= *(nums + i);
    }

    // XOR numbers from 1 to size-1
    for (int i = 1; i < size; i++)
    {
        result ^= i;
    }

    printf("Repeated element: %d\n", result);
}

int main()
{
    int size;

    printf("Enter size: ");
    scanf("%d", &size);

    int nums[size];

    printf("Enter array elements: ");
    for (int i = 0; i < size; i++)
    {
        scanf("%d", &nums[i]);
    }

    // Pass array using its address
    duplicate_element(nums, size);

    return 0;
}
```

Execution

